

CHEMISTRY

BY

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RESEARCH AND THEORY

THE ATOM AS A PART, VALENCE AS THE RULE, THE BOND LENGTH AS A FACT

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*For Megan, my loving wife, the only person that really understands
what I mean when I speak.*

*For Autumn and Opal, without you this thought would have never
occurred.*

To everyone else, I wish you well.

ABSTRACT

The subject is the building block and the rule. An atom is a point carrying values: a count of protons, a count of valence electrons, a radius, an electronegativity. A molecule or a solid is a set of such points in space. What tells that set from a random packing of the same atoms is one rule, valence, and valence fixes a bond to a length before anyone measures it. The engine this research paper is written for reads any domain as points carrying values and asks one question of the arrangement: how far it sits from the most disordered arrangement of its own parts. Chemistry is that question with the atom as the part.

What is quoted and what is new. The building blocks and the rules in the first four chapters are the settled content of general chemistry and are cited in the last: the periodic law, the shell counts, the octet, the electronegativity scale, the covalent and van der Waals radii, and the bond lengths tabulated from crystallography and gas-phase spectroscopy. Nothing about them is claimed here. What this research paper adds is their placement in the engine's six parts, and the predictions that placement makes. The atom is a representation, the octet is a sift, the disordered packing is a reference, and the bond length is an oracle of the strongest kind, a fact and not a consensus.

Nothing in this research paper has been measured here. The engine has readers for a protein and for a crystal, and both already rest on chemistry: a backbone repeats at three because three bond lengths are fixed, and a cell edge is a bond geometry tiled. It has no reader for a molecule itself, and the oracle README reserves the directory chemistry's bond lengths will occupy when they arrive. This research paper states what such a reader would find. The predictions are on the page before the reader exists. The one demonstration that runs today on the primitives already in the tree is in `examples/chemistry`, and it is named there as what it is.

Why no set here is bounded. The engine's own record is that every bound put on a quantity has had to be taken back off, and the reading improved each time. Chemistry invites the same error in a sharper form, because a bond either exists or it does not and a tolerance is the obvious way to decide. This research paper keeps to the engine's discipline instead: a departure is measured against a background drawn from the atoms themselves, and no threshold, no cutoff radius and no bounded element set is chosen by judgment. The question asked is always how far, never whether.

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THE ATOM IS A POINT CARRYING VALUES

1.1 ONE PART, SEVERAL MAGNITUDES

A part in chemistry is an atom, and an atom is not one number. It carries a count of protons Z , which names the element and orders the whole set; a count of valence electrons, the outer shell's occupancy; a radius, in two readings, the covalent one at which it bonds and the van der Waals one at which it merely touches; and an electronegativity [6], the pull it puts on a shared pair. Written together these are a vector, and every reading in this research paper is a magnitude taken from it or a distance taken between two of them. The engine reads any domain as points carrying values, and the atom is that description with the values fixed by the element.

Two atoms in a molecule are two such points at two positions, and the bond between them is the vector from one to the other. Its magnitude is the bond length. The representation has two parts: positions carrying element vectors, and the differences between positions carrying lengths. Nothing about it is particular to chemistry until the values are filled in. Hence the part of the engine that knows an atom exists is the only part that knows any domain exists.

1.2 THE ORDER IS A REAL READING, NOT A NUMBERING

The elements are ordered by Z [4], and that order is not a label chosen for convenience. Read a property along it, the atomic radius or the first ionization energy, and the sequence does not wander. It rises and falls with a period, and the period is the shell filling: two, then eight, then eight, then eighteen, then eighteen, then thirty-two. The periodic law [3] is the statement that a property of the atom is a periodic function of Z , and in the engine's terms it is the first departure chemistry offers. A property read along Z agrees with itself at the shell boundaries and disagrees with a shuffle of the same values.

The shell counts are themselves a departure from a flat expectation. A shell n holds $2n^2$ electrons because that is how many distinct states the quantum numbers allow it [5]. The capacities are 2, 8, 18, 32 and the period lengths follow from the order the sub-shells fill. This is not a count anybody chose. It is fixed before any element is weighed, the property the later chapters lean on: an answer known before the measurement and not supplied by it.

The element order, the electron set and the shell counts are the atomic structure, and this research paper reads them as given. It does not establish them and does not transcribe them. They are the ledger chemistry consumes, authored once where the atom is described, and everything this research paper adds is the bonding that sits on top of that ledger.

1.3 THE SET IS NOT BOUNDED

There is a temptation to write chemistry as a fixed table of some number of elements. The engine's record refuses it. Every bound put on a quantity in this work has had to be taken back off, and a bounded element set is the same defect waiting in a new place. The count of known elements is a fact about what has been synthesized, not about what the periodic law describes, and the law describes a sequence with no last term until a nucleus cannot hold together. A reading that fixes the set at a round number reads that number and not the chemistry.

The same holds one level down. An element is not one atom either. It is a set of isotopes differing in neutron count, weighted by abundance, and the weighting is a measured distribution and not a constant. Where a reading needs a mass it needs the distribution behind it, and a single tabulated value is a summary of that distribution with its spread deleted. The building block is a set, at every level at which it is read, and the engine's discipline is to draw the background from the set and not to cap it.

VALENCE IS A NECESSARY CONDITION

2.1 THE RULE THE SIFT ALREADY STATES

The engine's sift rests on one proposition: any subset of a pattern's points is a necessary condition. No selection rule built from the pattern can lose a true occurrence, and the converse fails for any proper subset. Every surviving candidate still has to be confirmed. Errors are one directional. A discrepancy is always an over-count, and it is detectable without knowing the answer.

Valence is a proposition of exactly that shape. A main-group atom in a stable molecule completes its octet [2, 1]: eight electrons in the outer shell, two for hydrogen and helium, counting shared pairs once for each atom that shares them. This is a condition on every atom at once, and a real molecule satisfies all of them. So the octet is a necessary condition on an arrangement, and a candidate structure that fails it at any atom is not a molecule. No true structure is ever refused by the octet, because a true structure meets it everywhere. What the rule discards is candidates, and that is cost, not correctness, the sift's own division carried onto chemistry unchanged.

2.2 A TRUTHY READING AT EACH ATOM

Read one atom and its bonds and the question is boolean. Sum the electrons it owns outright and the pairs it shares, compare against the shell it is filling, and the atom either closes or it does not. Carbon with four single bonds closes at eight and is true; carbon with three is a radical and is false; carbon with five cannot be drawn at all. The reading needs no tolerance. An electron pair is present or absent, a bond is drawn or not, and the count is an integer compared against an integer.

The molecule is the conjunction. It is true when every atom closes, and the sift's intersection is the operation that reads it: each atom's condition is a set of admissible local environments, and the arrangement survives where it lies in all of them at once. This is the same cascade the engine runs on a text or a lattice, with the anchor being an atom's satisfied octet instead of a rare symbol at an offset. Nothing is ported to bring it here. The sift was written over an index set with no interpretation, and an atom is one more thing an index can hold.

2.3 WHY THE EXACT COMPARE IS IRREDUCIBLE

The octet selects candidates and does not confirm them, exactly as the proposition warns. A drawn structure that satisfies every octet can still be the wrong isomer, because the same atoms with the same valences admit more than one arrangement: ethanol and dimethyl ether share a formula and

both close every atom. The rule narrows the field to the arrangements that could be molecules and leaves the choice among them to something the rule does not see, which for a real substance is its energy and for this engine is a departure measured against the disordered packing of the next chapter.

The one-directional error is worth stating in chemistry's terms. A structure the octet refuses is refused rightly: it cannot be a closed-shell molecule. A structure the octet admits is not thereby a molecule., a search that uses valence to prune loses nothing real and keeps some things that are not real, the only failure the sift permits and the reason a verification pass is never optional.

THE BOND LENGTH IS AN ORACLE

3.1 A FACT, NOT A CONSENSUS

The engine keeps its outside answers in one part, the oracle, and grades them. A family tree of languages is a reconstruction argued from cognates. Agreement with it is agreement with a scholarly consensus and where the two disagree nothing in the engine can say which is wrong. A published crystal cell edge is the other kind: somebody measured it, refereed it and wrote it down, and the instrument either recovers it or does not. The bond length is that second kind and more so. Valence fixes it whatever anyone believes. A carbon-carbon single bond is near 1.54 angstroms, a double near 1.34, a triple near 1.20, and a substance does not get a vote.

This is why the oracle README already reserves chemistry a directory. Bond lengths get their own place when they arrive, apart from the family trees, because a bond length is a fact and a family tree is a consensus and the two must not share a namespace. The value the engine consults for a protein already comes from this table: the backbone bonds are near 1.46, 1.52 and 1.33 angstroms, nitrogen to alpha carbon, alpha carbon to carbon, and the peptide carbon to the next nitrogen. None of those three was measured by the instrument that reads the backbone. They were the answer it was checked against.

3.2 TWO DOMAINS ALREADY REST ON IT

The engine has a reader for a protein and a reader for a crystal, and both stand on chemistry without naming it. The protein backbone is a chain of bonds whose lengths cycle with a period of three, because the three bond types repeat down the chain, and that period is fixed by chemistry before the chain is measured. It is the first positive control in the whole work with an answer not produced by the sample: a detector that returns anything other than three there is wrong, and no argument afterward can save it. The three came from valence, and valence is chemistry.

The crystal is the same fact tiled. A cell edge is a repeat distance, and what sets the repeat is the sizes of the atoms and the lengths of the bonds between them, packed into the smallest cell that reproduces under translation. The instrument recovers 453 of 453 published edges inside one voxel, and the thing it is recovering is a bond geometry stacked. So chemistry is not a third domain beside the protein and the crystal. It is the layer under both, the building blocks and the one rule that fix the lengths those two readers spend their time recovering.

3.3 THE FACT MAKES THE TRAP VISIBLE

An oracle that fixes a period exactly is also what exposes a detector's own failure. A sequence of period P agrees with itself as well at $2P$ and $3P$. A detector that takes the tallest lag reports a harmonic about as often as the period, and this has bitten three parts of the engine that did not know about each other. It was caught on the protein backbone precisely because chemistry said the answer was three: a reader returning six or nine could be called wrong on the spot. Without the fixed answer the harmonic passes as a clean result, a confident number with a factor of two that nothing flags.

That is the argument for chemistry as an oracle stated from the other side. It supplies an answer, and the answer is exact and external. An instrument checked against it cannot hide an error inside a tolerance. A bond length carries no spread that a detector's mistake can shelter in. It is the hardest test in the tree because the sample can argue with it least.

THE DISORDERED ARRANGEMENT OF THE SAME ATOMS

4.1 THE BACKGROUND IS DRAWN, NOT ASSUMED

The question the engine asks is always how far an arrangement sits from the most disordered arrangement of its own parts, and the most disordered arrangement is not a model to be posited. It is drawn from the object by deleting one property. Take the atoms a molecule holds, keep their counts exactly, and scatter their positions: the result is the least committal distribution consistent with the composition, the constrained maximum of entropy, and it sits at a single point because entropy is concave and a count constraint is linear. It carries no seed and no run-to-run variation, and a substance with no arrangement, its atoms placed at random, returns a departure of nothing.

Chemistry makes this concrete in a way a text does not. The composition is the formula, the shuffle is the same atoms at scrambled positions, and the departure is everything the bonds contribute over a gas of the same elements. A tolerance is the wrong tool here and it is a tempting one, because a bond either exists or it does not and a cutoff radius looks like the way to decide. The engine's record is that every such bound has had to come back off. So the reading is not whether two atoms are within some chosen distance. It is how far the observed set of distances sits from the set a random packing of the same atoms would give, with nothing chosen.

4.2 ORDER IS NOT AUTHORSHIP

There is a claim this construction had to give up once, and chemistry gives it up again from a new direction. The claim was that a departure from the disordered background measured human production: every corpus that departed had been made by a person. It fell when an unauthored sequence departed anyway. The gaps between primes return a departure outside the band every memoryless process occupies, and nobody wrote the primes.

A crystal is the same refutation in matter. A single-element metal packs into a lattice of long-range order, its atoms on a repeating grid recoverable to one voxel, and no person arranged it. The order is real, the departure is large, and the cause is physics minimizing an energy, not an author. So chemistry belongs with the primes and not with the prose: it is a domain where the arrangement is far from disordered and nothing produced it deliberately. A reading that took the departure as evidence of design would be wrong here on every mineral in the ground. The departure measures arrangement, and arrangement is not authorship, and the crystal is the cleanest place that difference can be shown.

4.3 WHICH SHUFFLE, AND WHAT IT ISOLATES

The engine grades its shuffles, and the grades transfer. A full permutation of the atoms destroys every arrangement at once and says only that arrangement is present, not at what scale. A shuffle that keeps local neighborhoods and destroys long ones separates a bond, which is local, from a lattice, which is not: the bond survives a long-range shuffle and the lattice does not. A sweep of the shuffle's reach locates where a structure lives. Scrambling within neighborhoods and keeping the coarse layout does the reverse, and separates the packing of a phase from the identity of the atoms packed.

None of these needs a distribution assumed for the disordered state, the whole point. A model of a random packing could be wrong. The atoms of the actual substance, with their positions deleted, cannot be, because it is the substance with one property removed and not a guess about what the substance is like. That is the sense in which no set here is bounded and no null here is derived: the null is the data, shuffled, and the only quantity reported is the distance to it.

WHAT A CHEMISTRY READER WOULD FIND

Nothing in this chapter has been measured here. It is the design chemistry adds to the engine and the predictions that design makes, stated before the reader exists. The engine reads a protein and a crystal but has no reader for a molecule itself, and the oracle for bond lengths is a directory reserved and not yet filled. The demonstrations that run on the primitives already in the tree are in `examples/chemistry` and are described as what they are; everything below is a prediction the reader would test.

Note, 26 September. The protein and crystal readers, `measure/periodicity` and the homes named below (`representation/atom`, `representation/structure`, `oracle/chemistry`) sat, or were to sit, under `src/python/`, which commit 74601c1 deleted on 26 September; none is in a file at orior 1948ae1. Four of the five scripts in `examples/chemistry`, `a_single_period_cannot_see_a_growing_one.py` among them, import from that tree. What this chapter says ran was run before the deletion. **Later, 26 September.** Commit d09b489 (orior PR 12) put back the 126 files 74601c1 deleted, each byte-identical to its state before the deletion, and biohub pins orior at d09b489. Every file named here that 74601c1 deleted is at the pin again; nothing here was rerun against them.

5.1 WHERE THE NEW PARTS GO, AND WHERE NOTHING IS ADDED

Two modules are new and the rest is reuse, and one thing chemistry needs is authored by another subject and only consumed here. The element itself, its proton count Z , its electron set, the Pauli exclusion [5] behind the shell counts, and the periodic recurrence, is the atomic structure, and that ledger is transcribed once in a shared `representation/atom` home and not copied under each subject that reads it. Chemistry consumes it. A duplicate element table under chemistry would be a second source of truth for a fact chemistry did not establish, and the engine holds a commit that would create one.

What chemistry adds is the bonding layer. A molecule needs a representation, atoms as points carrying element vectors and bonds as the vectors between them, which is a subject under `representation/structure` beside the protein and the crystal readers and is theirs to coordinate, since the three share what an atom is. The bond lengths need an oracle, `oracle/chemistry`, holding the tabulated lengths as facts apart from the language family trees. Nothing else is added. The octet runs on the sift's existing intersection, the disordered packing on the existing shuffles, and the departure on the existing measures. If the reuse is honest, no primitive learns that chemistry exists, and the whole subject lives in one representation directory and one oracle directory, over an element ledger it shares and does not own.

5.2 THE PREDICTIONS

First. The octet as a necessary condition will refuse no real molecule and will prune candidate arrangements, with the error one directional. A structure the octet admits may still be the wrong isomer. A verification pass stays mandatory, and the count of survivors will move with the composition exactly as the lattice cascade's does: a formula holding one element will over-admit because every atom is compositionally alike, and a formula holding several will prune hard.

Second. A bond-length oracle will be a positive control on par with the crystal cell edge and stronger than any family tree, because valence fixes the value and a substance cannot argue with it. A reader of molecular geometry checked against it will not be able to hide an error inside a tolerance, for the same reason the backbone period could be called wrong the moment it left three.

Third. A crystal of a single element will return a large departure from the disordered packing with nothing having authored it, placing chemistry beside the prime gaps and outside the band every memoryless process occupies. A reading that took departure as evidence of production will be wrong on every mineral, and this is the cleanest place in the tree to show that arrangement is not authorship.

Fourth. Reading a periodic property along Z will recover the shell structure only if the partition it is read against comes from outside the sequence. The engine reads a recurrence in two composed steps, and both fix a single period. `measure/periodicity` scores a candidate period against its own family of multiples, and a repeat at three is read as three and not as its harmonic six. `reference/periodic` then builds the maximum-entropy background at that one period, binding each position to the positions congruent to it modulo P . A periodic property along Z is not one period: the shell lengths are 2, 8, 8, 18, 18, 32, a sequence that grows. No single P makes the phase classes fall on the shell closures, and a reader restricted to one period reports a harmonic, a single wrong period, or nothing. The partition module names the fix. A partition is stipulated, estimated by a sweep, or supervised from outside. It is not stipulated, since a judgment-picked boundary set is the bounding this work forbids. It cannot be estimated, since a sweep for one period cannot recover a growing one, and that attempt is the trap. It is supervised: the shell boundaries are ground truth the atomic-structure ledger carries, the same kind of fact as the bond-length oracle and reached from a second direction. The prediction is that the real assignment departs far from its shuffle once the boundaries are supplied, and sits inside the shuffle band to a single-period reader without them. The method is shown to run in `examples/chemistry/4_measure/a_single_period_cannot_see_a_growing_one.py`, where a single-period reader departs on a constant recurrence and goes flat on a growing one while a reader handed the boundaries departs; that example writes no element data, and the real sequence and its boundaries are the ledger's. This is the harmonic trap, already met for one period in `measure/periodicity` and arriving again where the recurrence grows, named here so it cannot be met as a surprise.

Fifth. No reading here will need a bounded element set or a chosen cutoff radius, and any that appears to will be measuring the bound. The prediction is the engine's own history restated: the bound will have to come off, and the reading will improve when it does.

THE VALUE TABLES, NOT YET ENTERED

The specific magnitudes this research paper quotes are standard reference values and each needs its own verified entry: the covalent and van der Waals radii, the Pauling electronegativities, and the bond lengths, including the backbone values near 1.46, 1.52 and 1.33 angstroms the protein reader already consults. They are held in crystallographic and gas-phase spectroscopic tabulations and in the IUPAC recommendations, and the row for each stays empty here until it is opened and identified. Naming a value without the table it came from is the defect the citation tool exists to catch, and this chapter states the debt and does not hide it.

BIBLIOGRAPHY

Every measurement in this work is built on somebody's published result, and the first purpose of this chapter is that those people are credited by name. A surname in a footnote is not a citation. The second purpose is that the credit is checkable: each source is registered in `orior_citations`, where the copy sits itself, and this bibliography is generated from that registry and pinned to one commit of it. A reference therefore resolves not to a title but to exact bytes.

Credit here does not depend on agreement. Where this work reproduces a published result it says so, and where it fails to reproduce one it says that too, under the same names, because a refutation rests on the original as much as a confirmation does.

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